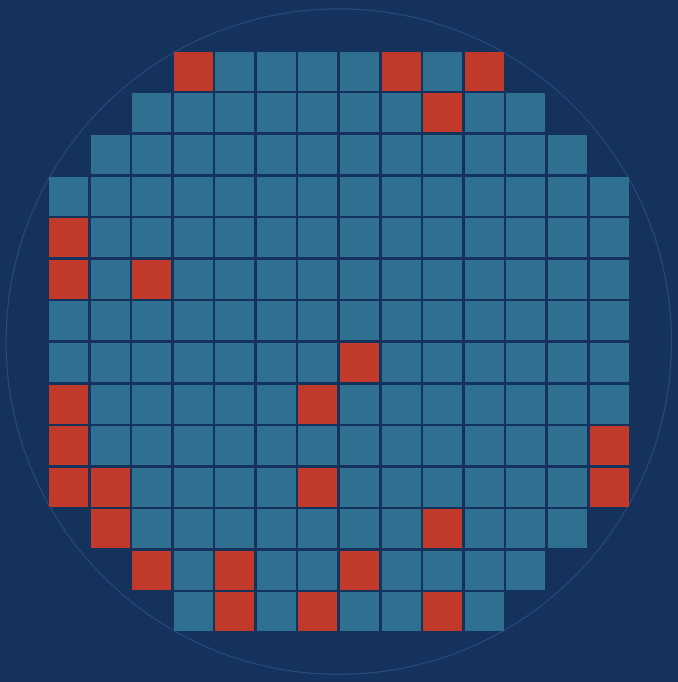
TECHNICAL DESIGN DOCUMENT · REV 1.0

AI-Optimized Yield Advisor

for Semiconductor Chip Testing

Wafer sort · Final test · System-level test



Document purpose. This document specifies the architecture, data model, AI methods, decision logic, deployment, and rollout plan for an AI-optimized Yield Advisor that sits across wafer sort (CP), final test (FT), and system-level test (SLT). It is written for test, product, yield, and data engineering leads who will review, build, and operate the system.

Item	Value
Scope	Probe (CP1/CP2), final test (FT hot/cold/room), SLT, burn-in correlation
Primary users	Yield engineers, test engineers, product engineers, quality, test-floor operations
Deployment	Hybrid: tester-cell edge agents + central lakehouse (on-prem or VPC)
Key outcomes	+1 to 3 pts sort yield recovered, 20 to 40% faster excursion detection, 8 to 15% test time reduction, DPPM held or reduced
Status	Design for review. Targets are planning estimates to be validated in the pilot (Section 13).

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1. Executive Summary

Semiconductor test generates terabytes of per-die parametric and functional data every week, yet most of it is consumed only as pass/fail bins and weekly yield charts. Yield loss is found late, root cause takes days of manual commonality analysis, test limits are static compromises between overkill and escapes, and test programs carry redundant items that cost tester time on every unit.

The Yield Advisor turns that data into ranked, explainable, actionable recommendations. It combines five AI engines over a shared die-level data foundation:

- **Wafer Map Pattern AI** classifies spatial failure signatures (edge ring, scratch, center, cluster and others) on every wafer within seconds of sort completion and links each signature to its likely process or probe cause.
- **Excursion Detector** flags yield and parametric shifts with EWMA/CUSUM and multivariate anomaly models, catching small sustained shifts that Shewhart charts miss.
- **Root Cause Engine** ranks equipment, chamber, recipe, probe card, tester, and material factors by their contribution to the loss using gradient-boosted models with SHAP attribution and classical commonality statistics.
- **Adaptive Limits and Outlier Screening** applies Dynamic Part Average Testing (DPAT), Good-Die-in-Bad-Neighborhood (GDBN/NNR), and multivariate Mahalanobis screens to remove latent-defect die that pass specification limits.
- **Test Time Optimizer** identifies redundant or low-coverage test items and proposes sampling or removal under an explicit, statistically bounded escape risk.

An **Advisor layer** scores every candidate action by expected yield or cost benefit, confidence, and quality risk, then presents it with evidence. An LLM copilot grounded with retrieval over historical 8D reports, failure analysis, and test program documentation explains findings in plain language and drafts change requests. No test program, limit, or disposition changes without human approval through the existing change control process.

> **Bottom line:** for a product line running 10,000 wafers a month, recovering 1.5 yield points and cutting 10% of test time is worth on the order of USD 9 to 10 million per year (worked example in Section 12), with a 12-month phased rollout starting from a read-only pilot.

2. Problem Statement and Scope

2.1 Where yield is lost in test

Stage	Typical loss mechanisms	Pain today
Wafer sort (CP)	Process defects, parametric drift, probe contact and card wear, prober temperature, edge die	Spatial patterns reviewed manually for a sample of wafers; probe vs. process loss confused
Final test (FT)	Assembly damage, package stress, thermal and speed binning loss, handler contact	Correlation to sort data is manual; retest policy fixed rather than data-driven
System-level test	Marginal timing, power integrity, firmware interaction	Fails discovered late and expensive; weak link back to die-level history
Field / RMA	Latent defects that passed all test	Escapes traced back by hand; screening rules added reactively

2.2 Goals

- Detect yield excursions within one lot of onset rather than at the weekly review.

- Cut mean time to root cause (MTTRC) from days to hours for the top-loss signatures.
- Recover yield lost to overkill (good die rejected) without raising escapes.
- Reduce DPPM through statistically grounded outlier screening.
- Reduce test time per unit through evidence-based test content optimization.
- Give every recommendation a quantified benefit, a confidence, and a traceable evidence trail.

2.3 Non-goals

- Autonomous changes to production test programs. The system recommends; humans release.
- Replacing fab-side defect inspection or process control systems. The Advisor consumes their data.
- Design-for-test (DFT) pattern generation. ATPG remains in the EDA flow; the Advisor supplies feedback on pattern effectiveness.

3. Yield Fundamentals Used by the System

The Advisor's quantitative reasoning rests on standard yield definitions so that its outputs reconcile with existing yield reports.

3.1 Definitions

Metric	Definition
Wafer sort yield	Y_{sort} = good die / gross die per wafer (GDPW), per wafer, lot, and product
Line-to-ship yield	$Y_{total} = Y_{line} \times Y_{sort} \times Y_{assy} \times Y_{FT} \times Y_{SLT}$
Systematic vs. random	$Y_{sort} = Y_{sys} \times Y_{rand}$. Spatial patterns and parametric shifts drive Y_{sys}; distributed defects drive Y_{rand}
Overkill	Good die rejected by test (false fail). Estimated from retest recovery, guard-band analysis, and downstream correlation
Underkill (escape)	Defective die passed by test. Measured as DPPM from downstream fails and field returns
Test escape DPPM	DPPM = escapes / units shipped $\times 10^6$
Retest recovery rate	Share of first-pass fails that pass on retest, by bin. High rates indicate contact or limit issues

3.2 Random-defect yield models

For a die of critical area A and defect density D_0, the Advisor fits the negative binomial model, which reduces to Poisson as the clustering parameter grows:

$$Y_{rand} = (1 + A\,D_0 / \alpha)^{-\alpha}$$

Fitting D_0 and α per wafer from the non-patterned fail density lets the system separate the random component from systematic loss. That split decides who owns the recovery: the fab (defect density) or test and product engineering (systematic and test-induced loss).

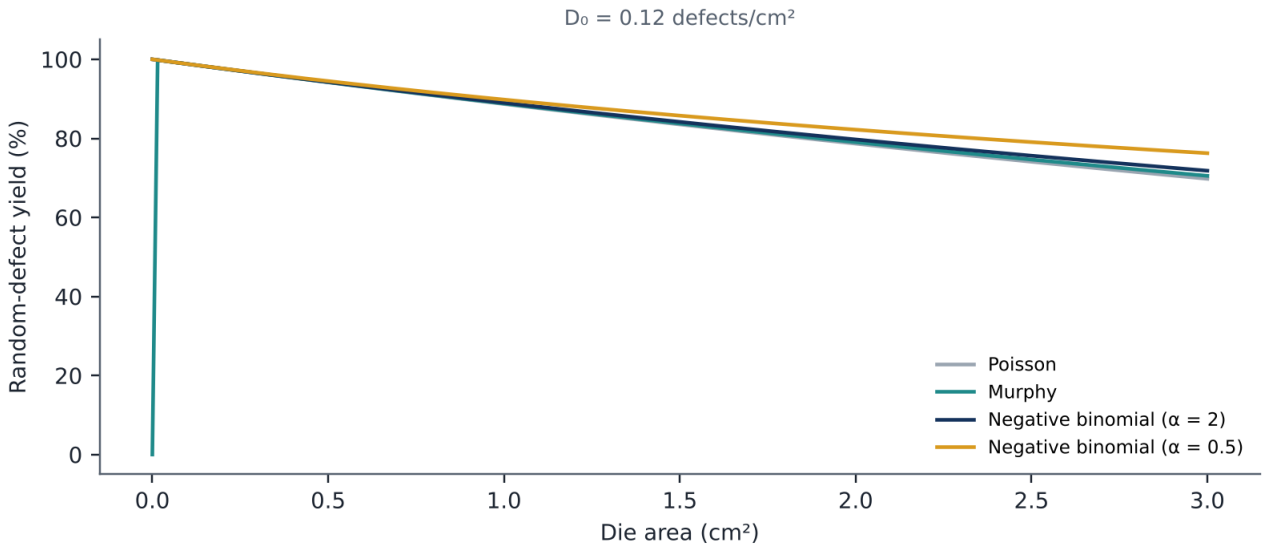


Figure 1. Random-defect yield vs. die area under common models. Lower α means stronger defect clustering and higher yield for the same D_0.

4. System Architecture

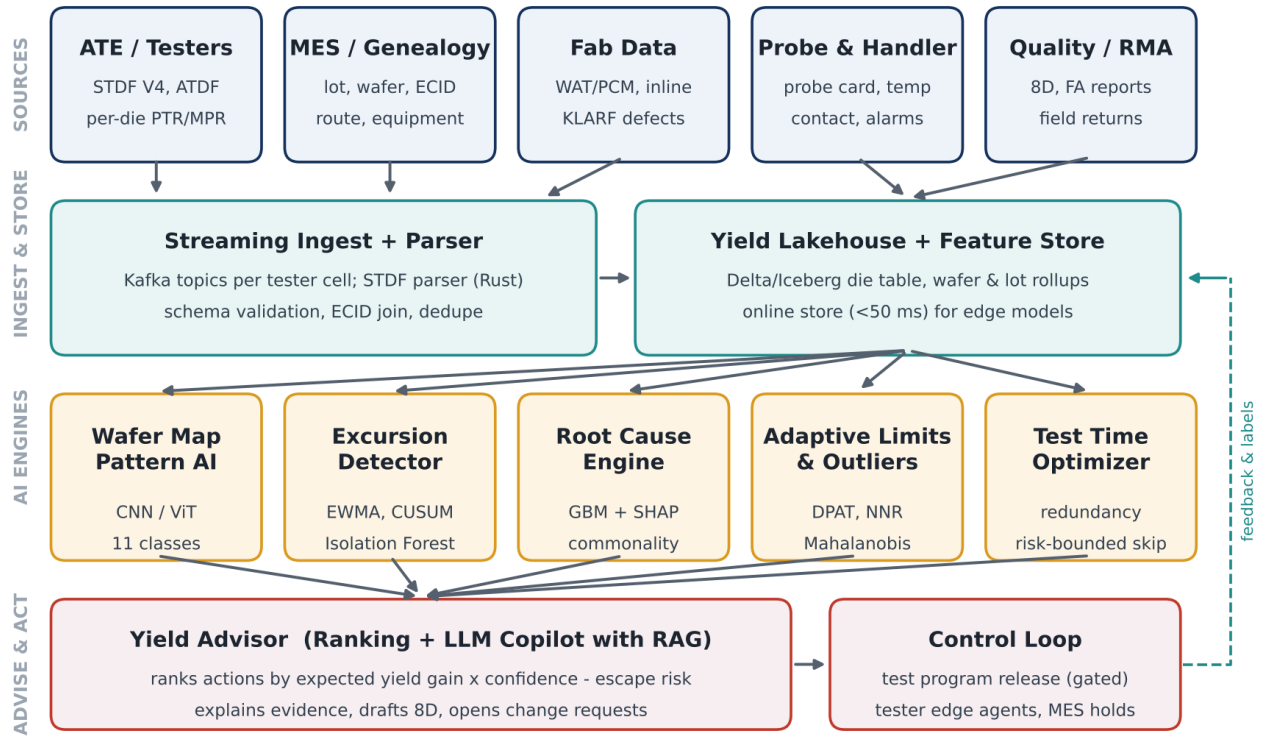


Figure 2. Logical architecture. Data flows from test and fab sources through streaming ingest into a lakehouse and feature store, feeds five AI engines, and converges on the Advisor, which drives a gated control loop.

4.1 Layers

Layer	Responsibility	Reference technology
Edge agent (per tester cell)	Tail STDF as it is written, parse, push to stream; run low-latency models (DPAT, NNR, real-time contact alarms)	Rust STDF parser, ONNX Runtime, gRPC
Streaming	Durable event bus, one topic per site and stage, exactly-once delivery to storage	Apache Kafka or Redpanda
Lakehouse	Die, wafer, lot, and test-item tables with time travel for reproducible analysis	Delta Lake or Apache Iceberg on object storage; Spark / Trino
Feature store	Offline features for training, online features under 50 ms for edge scoring	Feast or equivalent; Redis online store
Model serving	Batch scoring per wafer and lot, online scoring for edge models	Ray Serve or KServe; ONNX
Advisor service	Recommendation scoring, evidence assembly, LLM copilot, workflow integration	Python service, vector index, LLM API
Presentation	Yield cockpit, wafer map viewer, recommendation inbox, APIs	Web UI, REST/GraphQL, MES and ticketing connectors

4.2 Latency tiers

Tier	Budget	Examples
Real time	< 1 s per die	Contact resistance drift, consecutive-fail stop, NNR on the fly
Near real time	< 2 min after wafer end	Wafer map classification, DPAT final limits, wafer disposition advice
Lot	< 15 min after lot end	Excursion detection, retest recommendation, lot hold advice
Analytical	Hours to daily	Root cause ranking, test time optimization, model retraining

5. Data Foundation

Model quality is bounded by data joins. The most important design decision is a die-level record keyed by a stable identity that survives from wafer sort through final test and into the field.

5.1 Sources

Source	Content	Format / cadence
ATE test results	Per-die parametric (PTR), multi-result (MPR), functional (FTR) results, bins, test times	STDF V4 per wafer or lot; streaming
MES and genealogy	Lot history, route, equipment, chamber, recipe, timestamps, splits and merges	Database / API; event-driven
Fab metrology	WAT/PCM e-test, inline CD and film, defect inspection maps	CSV, KLARF; per lot
Probe / handler logs	Probe card ID and touchdowns, cleaning events, chuck temperature, handler site, jams	Equipment logs, SECS/GEM
DFT diagnosis	Scan fail logs, diagnosed suspect nets and cells	Diagnosis tool reports
Quality	8D reports, failure analysis, RMA and field returns	Documents, QMS records

5.2 STDF V4 records consumed

Record	Name	Use in the Advisor
MIR / MRR	Master Information / Results	Lot, product, tester, program revision, start/end time
WIR / WRR / WCR	Wafer Information / Results / Configuration	Wafer ID, die counts, flat orientation, coordinate system
PIR / PRR	Part Information / Results	Die X/Y, site, hard and soft bin, test time, pass/fail
PTR	Parametric Test Record	Per-test value, limits, units: the main input to DPAT and modeling
MPR	Multiple-Result Parametric	Pin-level results (leakage, continuity) for pin-pattern features
FTR	Functional Test Record	Vector fail counts and fail pins for scan and memory signatures
HBR / SBR	Hard / Soft Bin Records	Bin definitions and summary counts for reconciliation
TSR	Test Synopsis	Per-test execution counts and times for test time optimization

5.3 Die identity and genealogy

Each die carries a canonical key `(fab, lot, wafer, x, y)`. Where the design includes an electronic chip ID (ECID) fuse, it is read at sort and at final test so that FT, SLT, and field results join back to the exact wafer location. Without ECID, the system uses assembly lot traceability and reports joins as probabilistic, with that uncertainty carried into any downstream model.

5.4 Core tables

```
die_result     (die_key, stage, insertion, temp, site, hbin, sbin, pass, test_time_ms,
                tester_id, probe_card_id, program_rev, ts)
test_value     (die_key, stage, insertion, test_num, test_name, value, lo_lim, hi_lim, unit)
wafer_summary  (wafer_key, stage, gross, good, yield, pattern_class, pattern_conf, d0, alpha)
lot_context    (lot_key, step, equipment_id, chamber, recipe, ts_in, ts_out)
recommendation (rec_id, type, scope, expected_gain, confidence, risk, evidence_uri,
                status, owner, created_ts, decided_ts)
```

Volume planning: a 10,000 wafers per month line at 600 GDPW and 2,000 parametric tests per die produces about 12 billion test values a month. Columnar storage with dictionary encoding brings this to roughly 1.5 to 3 TB per month for the test_value table.

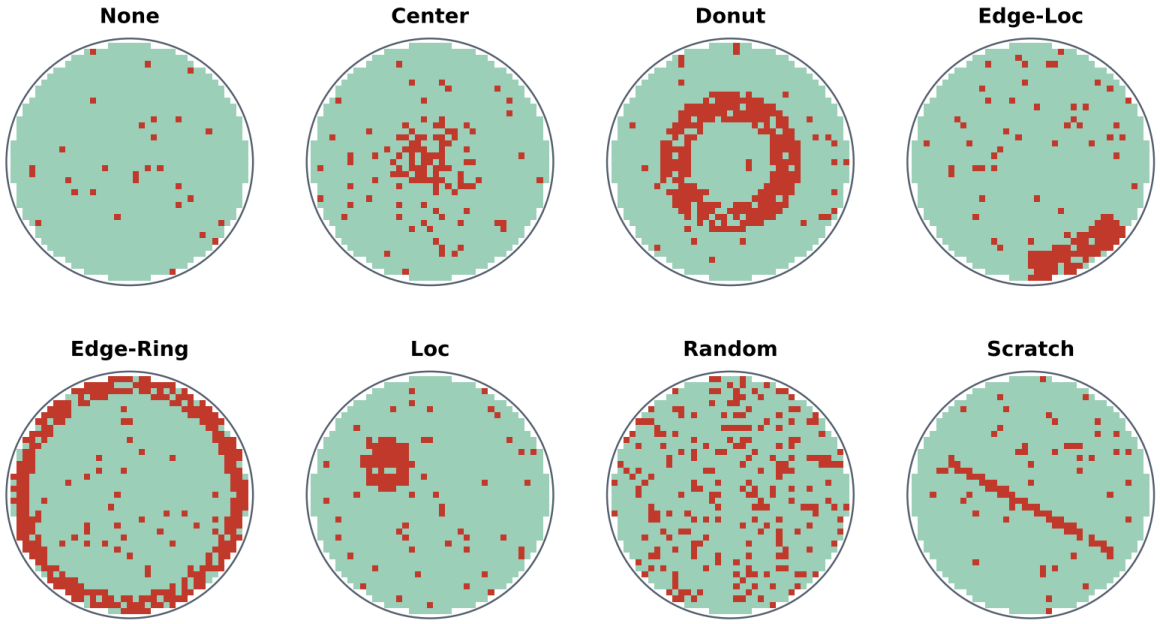
5.5 Data quality gates

- Schema and record-order validation for every STDF file; files with missing PRR or truncated records are quarantined.
- Limit consistency: program revision and limits are versioned; a value is never compared against limits from a different revision.
- Coordinate normalization to a common notch-down orientation before any spatial model.
- Duplicate and retest handling: every insertion is kept, flagged by insertion number, and a final-disposition view is derived.
- Completeness monitors per tester and site so that missing data is never mistaken for a yield change.

6. AI Engines

6.1 Wafer Map Pattern AI

Task. Classify each wafer's pass/fail map (and optionally per-bin maps) into failure signature classes, localize the pattern, and estimate the die lost to it.



Pass (green) / fail (red) die per wafer, synthetic examples of the WM-811K defect classes

Figure 3. Representative spatial signatures. Each class maps to a family of likely causes that the Root Cause Engine then tests.

Signature	Typical causes	First check
Edge-Ring	Etch or deposition uniformity, bevel issues, edge bead removal	Process chamber, edge exclusion
Center	CMP non-uniformity, spin-coat thickness, thermal center hot spot	CMP tool, track, anneal
Donut	Thermal or plasma ring non-uniformity	Chamber heater zones, RF
Edge-Loc / Loc	Handling contact, localized particles, chuck contamination	Wafer handler, chuck
Scratch	Mechanical handling, CMP slurry agglomerates	CMP pad, robot end effector
Repeating reticle	Mask defect or stepper field issue	Reticle ID, exposure field map
Probe-induced	Bad needle, card planarity, site-specific contact	Probe card, site-to-site yield

Model. A convolutional network (ResNet-18 scale) on maps resampled to 64 x 64, with a three-channel input (fail, bin class group, die-valid mask). A vision transformer variant is evaluated in the pilot. Mixed-type wafers (two patterns at once) are handled as multi-label with sigmoid outputs; a segmentation head (U-Net) outlines the pattern region so lost die can be counted.

Training data. Public WM-811K (811,457 maps, of which about 172,950 are labeled into 9 classes) and MixedWM38 for pre-training, then fine-tuned on in-house labeled wafers. Class imbalance (the majority are 'none') is handled with focal loss, rotation and flip augmentation (patterns are rotation-equivariant), and synthetic pattern injection.

Open-set handling. New signatures must not be forced into known classes. An energy-score out-of-distribution detector routes low-confidence wafers to an 'unknown' queue; embeddings are clustered weekly (HDBSCAN) so engineers can label new classes.

Targets. Macro F1 at or above 0.93 on in-house hold-out, at least 0.85 recall on each rare class, classification under 200 ms per wafer on CPU.

6.2 Excursion Detector

Task. Detect statistically meaningful shifts in yield, bin rates, and key parametric distributions at the earliest possible lot.

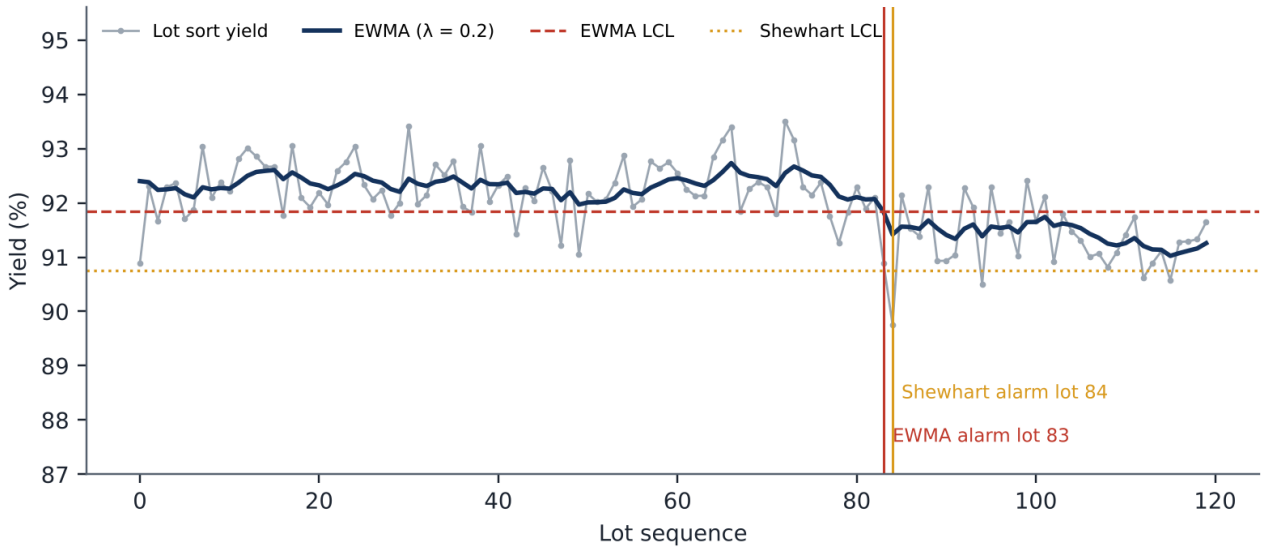


Figure 4. A sustained shift of about 1.7σ begins at lot 78. EWMA accumulates evidence across lots; a Shewhart chart reacts only to single extreme points. On real data EWMA typically signals small shifts several lots earlier.

Methods. The detector layers several complementary checks:

- Univariate EWMA and CUSUM charts on yield, each top soft bin, and the median and spread of critical parametrics, by product, tester, and probe card.
- Multivariate monitoring with Hotelling T^2 on PCA-reduced parametric signatures per wafer, catching shifts spread across many tests that are individually in control.
- Isolation Forest on wafer-level feature vectors (pattern probabilities, bin mix, parametric moments) for novel anomalies.
- Change-point detection (PELT) for retrospective dating of when a shift began, which narrows the root cause search window.

False alarm control uses Benjamini-Hochberg across the many simultaneous charts, so engineers are not flooded. Alarm thresholds target an in-control average run length of about 500 lots per chart family.

6.3 Root Cause Engine

Task. Given an excursion or a persistent loss signature, rank the factors most likely responsible.

Commonality analysis. For every process step, compare the distribution of affected vs. unaffected wafers across equipment and chambers using chi-square or Fisher exact tests, and Kruskal-Wallis for continuous responses. This is the classical method engineers already trust and it anchors the ranking.

Model-based attribution. A gradient-boosted tree model (LightGBM) predicts wafer yield loss or signature probability from routing, equipment, timing (queue times), WAT, and probe context. SHAP values give each factor's contribution per wafer and in aggregate. Interaction values expose combinations such as 'chamber B only after a PM'.

Causal guardrails. Equipment assignment is confounded with time and product mix. The engine adjusts for these with stratification and, for important decisions, runs a double machine learning estimate of the effect of a suspected tool with time and product as controls. Each ranked factor is reported with effect size, p-value, wafers affected, and the time window, and is labeled 'association' unless a split-lot or tool-down experiment confirms it.

Output example.

Rank	Factor	Evidence	Est. loss	Confidence
1	Etch ETCH-07 chamber C, after PM on 9 Oct	Edge-Ring rate 31% vs 4% baseline; Fisher p < 1e-9; SHAP top	1.8 pts	High
2	Probe card PC-4412 site 3	Site 3 yield 3.1 pts below other sites; contact Rc trend rising	0.6 pts	High
3	WAT Vt_n shift +18 mV on lots 2214-2229	Correlates with SB07 Vmin (r = 0.62)	0.4 pts	Medium

6.4 Adaptive Test Limits and Outlier Screening

Task. Remove die that pass specification limits but are statistically abnormal (latent reliability risk), and right-size guard bands to cut overkill.

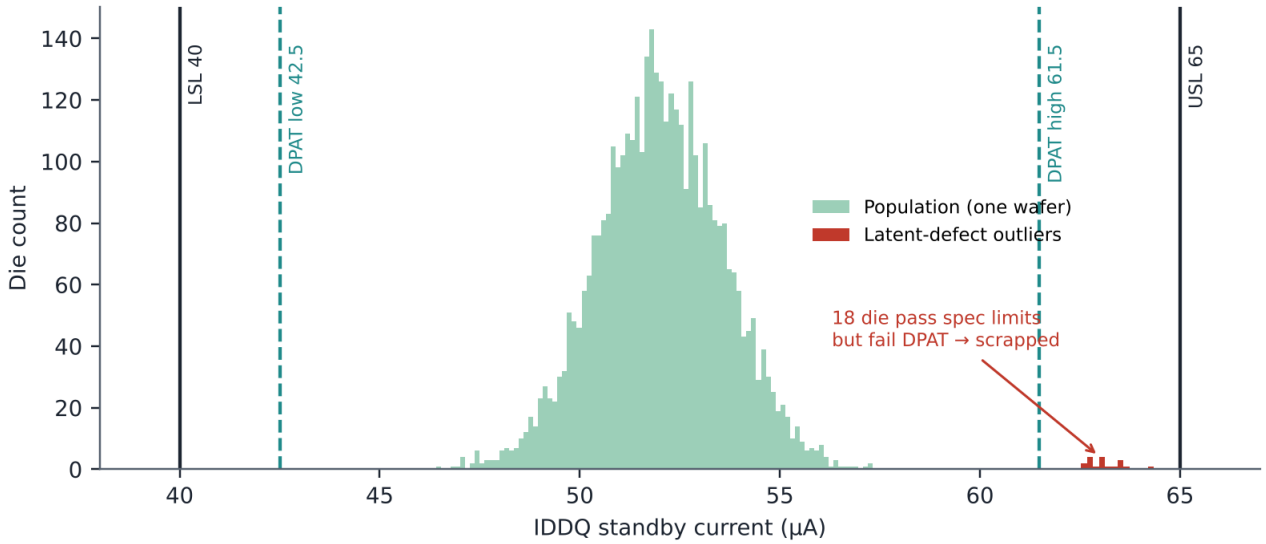


Figure 5. DPAT on one wafer. Limits computed from the wafer's own robust distribution flag outliers that sit well inside the specification limits.

Part Average Testing. Following the AEC-Q001 guideline, static PAT limits are set at a robust mean plus or minus 6 robust sigma from qualification data; dynamic PAT recomputes them per wafer or lot:

```
robust_mean  = median(x)
robust_sigma = (Q3 - Q1) / 1.35
DPAT limits  = robust_mean +/- k * robust_sigma      (k = 6 default, tuned per test)
apply only if within spec:  max(LSL, low)  ...  min(USL, high)
```

- **Good Die in Bad Neighborhood (GDBN).** A passing die surrounded by failing die carries elevated latent defect risk. The engine computes a weighted neighbor fail score and also applies

cluster-based rules for die inside or bordering a detected pattern region.

- **Nearest Neighbor Residual (NNR).** For spatially trending parametrics such as IDDQ, a die's value is compared with the median of its neighbors rather than with the whole wafer, which catches outliers that wafer-level PAT hides.
- **Multivariate PAT.** Robust Mahalanobis distance (Minimum Covariance Determinant) across correlated tests, for example IDDQ at several voltages, catches die normal on each test but abnormal in combination.
- **Guard-band optimization.** Using retest data and downstream correlation, the engine estimates measurement repeatability per test and proposes guard bands sized to the measured gauge uncertainty instead of legacy fixed margins. This is the main lever for recovering overkill.

Calibration to quality. Every screen is evaluated on yield cost and DPPM benefit, using burn-in and field return labels where available. A screen is recommended only if its expected DPPM reduction per yield point lost exceeds the product's quality-cost threshold.

6.5 Test Time Optimizer

Task. Reduce tester seconds per unit without raising escapes.

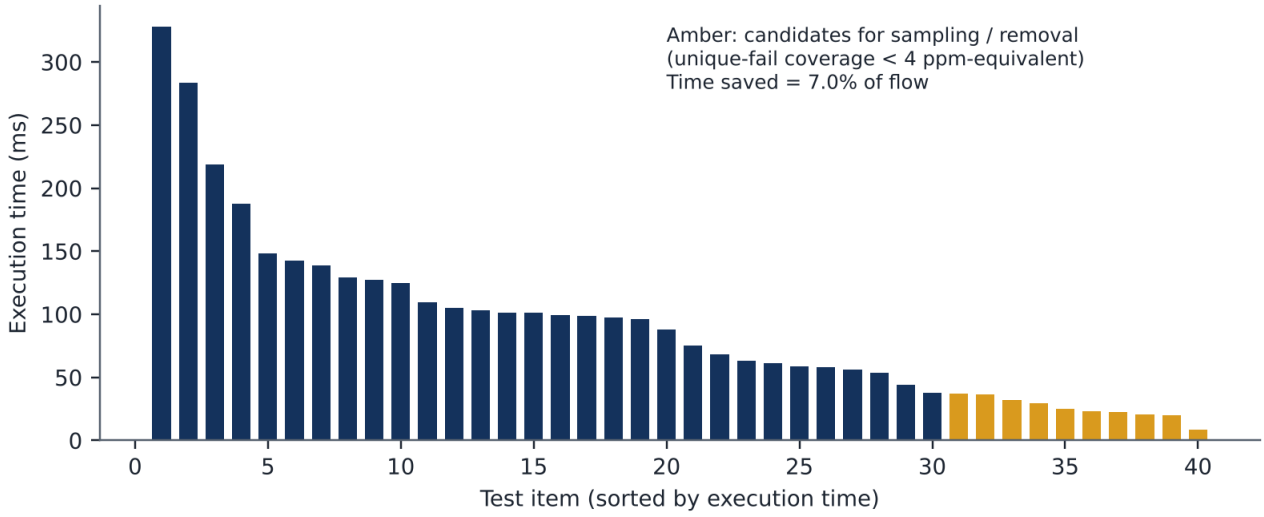


Figure 6. Candidate test items for sampling or removal in an example flow. Removal is proposed only when unique-fail coverage is below the risk bound.

- **Redundancy analysis.** For each test, count 'unique fails': die that fail only this test. Tests with near-zero unique fails over millions of units are candidates. Correlated parametrics ($r > 0.98$) suggest one test can predict another.
- **Risk bound.** With zero unique fails in n units, the 95% upper bound on the escape rate from removing a test is about 3/n (rule of three). Removal is proposed only when this bound is below the DPPM budget allocated to test optimization.
- **Adaptive test.** A per-die model predicts pass probability from early tests and sort history; low-risk die skip expensive items while a random audit sample (for example 2%) always runs the full flow to keep the model honest.
- **Test reordering.** Tests with high fail probability per second run earlier so that failing die stop sooner under stop-on-fail.

6.6 Yield Prediction

A per-lot model predicts sort yield and bin mix from WAT and inline data before the lot reaches probe. This supports capacity planning and lets the Advisor flag lots likely to excurse so the test floor can prepare extra sampling. Model: gradient boosting with quantile outputs (P10, P50, P90); target mean absolute error under 1.2 yield points.

6.7 Model summary

Engine	Primary model	Inputs	Key metric (target)
Wafer map	CNN / ViT + U-Net, OOD energy score	Die pass/fail and bin maps	Macro F1 >= 0.93
Excursion	EWMA, CUSUM, Hotelling T2, Isolation Forest	Lot and wafer KPIs	Detect 1.5σ shift within 3 lots
Root cause	LightGBM + SHAP; commonality tests; DML	Genealogy, WAT, probe context	True cause in top 3 >= 80%
Outliers	DPAT, NNR, GDBN, robust Mahalanobis	Per-die parametrics, neighbors	DPPM down at < 0.3 pt yield cost
Test time	Unique-fail analysis, adaptive test classifier	Per-test results, TSR times	8 to 15% time cut, no escape increase
Yield forecast	Quantile gradient boosting	WAT, inline, history	MAE < 1.2 pts

7. Advisor Layer: Recommendation Logic

Each engine emits candidate actions. The Advisor normalizes them into one inbox ranked by risk-adjusted value, so a yield engineer sees the most valuable decision first, whatever engine produced it.

7.1 Scoring

```
score = P(correct) x expected_benefit_usd  -  expected_quality_cost_usd  -  effort_cost_usd

expected_benefit_usd   = recovered_die/month x die_value  +  tester_sec_saved x cost_per_sec
expected_quality_cost  = delta_DPPM x units x cost_per_escape   (escape cost set by customer tier)
P(correct)             = calibrated model confidence (isotonic regression on past outcomes)
```

Confidence is calibrated against the outcome of previous recommendations, so a stated 80% means roughly four in five similar recommendations were confirmed. Actions that touch shipped quality (limit widening, test removal) also need the DPPM bound check from Section 6.5 to pass before they can appear.

7.2 Action catalogue

Action type	Example	Approval
Investigate	Open a case for Edge-Ring rise on ETCH-07 C with evidence pack	Yield engineer
Equipment	Recommend chamber inhibit pending PM check	Process owner via MES
Probe	Clean or replace probe card PC-4412; adjust touchdown count limit	Test floor lead
Retest	Retest SB30 contact fails on lot 2231 (expected recovery 72%)	Auto within policy, else test engineer
Disposition	Ink 18 DPAT outliers; GDBN ink ring around cluster	Quality engineer
Limit change	Tighten IDDQ PAT k from 6 to 5 for automotive grade	Change control board
Test content	Move TEST_4120 to 5% sampling (bound 0.6 DPPM)	Change control board

7.3 LLM copilot

A language model makes the evidence accessible: engineers ask questions such as 'why did lot 2214 drop on SB07?' and receive an answer assembled from engine outputs, charts, and similar past cases.

- **Grounding.** Retrieval over 8D reports, FA results, test program release notes, and specification documents, indexed by product and signature. Every claim in an answer links to its source record.
- **Tool use.** The copilot calls the same query APIs as the UI (yield by factor, wafer maps, commonality runs). It never calculates statistics itself; numbers come from the engines.
- **Drafting.** It drafts 8D D2 to D4 sections, change request justifications, and shift handover summaries for human editing.
- **Guardrails.** Read-only data access, no direct write path to test programs or MES, refusal when evidence is insufficient, and logging of every prompt and response for audit.

8. Human-in-the-Loop and Governance

Test programs and limits are controlled quality documents, and in automotive and medical markets they are tied to customer-approved control plans. The Advisor fits that process rather than bypassing it.

- **Autonomy levels.** Level 0 (inform), Level 1 (recommend, human executes), Level 2 (auto-execute within a pre-approved policy, such as retest of contact bins up to a cap). Production starts at Level 1; only explicitly approved policies run at Level 2.
- **Change control.** Limit and content changes generate a change request with evidence pack, risk bound, and rollback plan, routed to the existing change control board. Automotive products follow PPAP and customer notification rules.
- **Traceability.** Every recommendation stores model version, data snapshot, and decision, so any shipped unit's test conditions can be reconstructed.
- **Feedback.** Accept, reject, and outcome labels feed retraining and confidence calibration. Rejection reasons are mandatory picklists so the system learns why.

9. MLOps and Model Lifecycle

Stage	Practice
Versioning	Data snapshots via lakehouse time travel; models, features, and thresholds in a registry (MLflow)
Validation	Hold-out by time (never random split) to mimic deployment; per-product and per-tester slices; rare-class recall gates
Shadow mode	New models run beside production for at least 4 weeks; recommendations compared but not shown
Release	Staged by product family; automatic rollback if acceptance rate or calibration degrades
Drift monitoring	Population stability index on key features; prediction distribution and OOD rate per model; alerts on program revision changes
Retraining	Scheduled monthly plus event-triggered on new product, new program revision, or drift alarm
Explainability	SHAP summaries, Grad-CAM overlays on wafer maps, evidence links in every recommendation

New-product introduction is the hardest case because no history exists. The system uses transfer learning from sibling products for wafer map models, falls back to commonality statistics and DPAT (which need no training) for the first lots, and switches to product-specific models once enough wafers are labeled.

10. Deployment and Integration

10.1 Topology

- **Edge.** One agent per tester cell or per group of testers, on a small server on the test floor network. It never blocks the tester: if the agent is down, testing continues and data backfills.
- **Central.** Lakehouse, model training, and Advisor services in the company's data center or a dedicated cloud VPC. Multi-site deployments keep raw data in region and replicate only aggregates when data residency requires it.
- **OSAT and foundry data.** Partners deliver STDF through secured transfer; a partner-facing ingest validates and maps it to the canonical schema.

10.2 Integrations

System	Direction	Purpose
ATE (Advantest, Teradyne and others)	In	STDF stream; optional real-time API for stop-on-alarm
Prober / handler	In	Temperature, touchdowns, alarms (SECS/GEM)
MES	In / out	Genealogy in; hold and inhibit recommendations out through approved transactions
Inking / map server	Out	Updated wafer maps with outlier inks after approval
Existing yield tools	In / out	Coexists with tools such as Exensio, yieldWerx, or in-house YMS; exchanges signatures and recommendations
Ticketing / QMS	Out	Cases, change requests, 8D drafts

10.3 Performance and sizing (10k wafers per month)

Component	Sizing
Ingest	Peak 20 MB/s STDF across 60 testers; 3-broker Kafka cluster
Storage	about 3 TB/month compressed, 24-month hot retention, archive after
Compute	Spark cluster 16 to 32 vCPU for daily jobs; 1 GPU for wafer map training; CPU inference
Availability	Edge 99.9% (non-blocking); Advisor 99.5%; RPO 1 h, RTO 4 h

11. Security and IP Protection

- Test data reveals design and process IP. All data is encrypted in transit (TLS 1.3) and at rest (AES-256) with customer-managed keys.
- Role-based and attribute-based access control: product teams see their products; partners see only their own lots.
- Fabless and OSAT multi-tenancy uses separate storage namespaces and key per tenant; cross-tenant learning uses only model weights trained on approved data or federated updates.
- The LLM copilot runs on an enterprise endpoint with no training on customer prompts; retrieval respects the same access controls as the UI.
- Audit logging of all data access, recommendation decisions, and copilot sessions, aligned with ISO 27001 and, for automotive, TISAX expectations.

12. KPIs and ROI Model

12.1 KPIs

KPI	Baseline (typical)	Target after 12 months
Excursion detection latency	3 to 7 days (weekly review)	< 1 lot / < 24 h
Mean time to root cause	3 to 10 days	< 1 day for known signatures
Wafers with classified signature	Sample-based manual review	100%, automatic
Overkill (retest-recoverable loss)	1 to 3% of fails	Halved
Test time per unit	Baseline	-8 to -15%
Escape DPPM	Baseline	Equal or lower, never higher
Recommendation acceptance rate	n/a	> 60%, with confirmed-benefit rate > 70%

12.2 Worked example (illustrative)

Assumption	Value
Wafers per month	10,000
Gross die per wafer	600
Value per good die	USD 8
Sort yield recovered	1.5 points
Units through final test per month	5.4 million
Test time per unit / tester cost	4.0 s / USD 0.012 per second
Test time reduction	10%

Benefit line	Calculation	Per month	Per year
Yield recovery	10,000 x 600 x 1.5% x USD 8	USD 720,000	USD 8.64 M
Test time	5.4 M x 0.4 s x USD 0.012	USD 25,920	USD 0.31 M
Engineering time	approx. 4 FTE of analysis effort redeployed	USD 50,000	USD 0.60 M
Total		**USD 795,920**	**USD 9.55 M**

Against an estimated first-year cost of USD 1.5 to 2.5 million (team, infrastructure, licenses), payback falls within the first two to four months of full deployment. Escape-cost avoidance from outlier screening is excluded from the total because it depends heavily on market; for automotive products it is often the largest benefit.

13. Implementation Roadmap

Phase	Duration	Scope	Exit criteria
0. Foundation	Months 1-2	STDF ingest from 1 site, die table, genealogy join, data quality gates, yield cockpit	> 99% of wafers ingested within 10 min; joins validated against YMS
1. Pilot (read-only)	Months 3-5	Wafer map AI, excursion detector, commonality RCA on 2 product families; shadow recommendations	Macro F1 >= 0.90; 3 confirmed root causes found faster than baseline
2. Advise	Months 6-8	Advisor inbox, LLM copilot, DPAT/NNR recommendations, retest advice, ticketing integration	Acceptance > 50%; measured yield gain > 0.5 pt on pilot products
3. Optimize	Months 9-12	Guard-band and test time optimization through change control; Level 2 retest policy; multi-site rollout	Test time -8% on one product with no DPPM increase over 3 months
4. Scale	Year 2	All products, OSAT data, yield forecasting, federated cross-site learning	Program KPIs in Section 12 met

13.1 Team

Core team of 8 to 10: product owner (senior yield engineer), 2 data engineers, 2 ML engineers, 1 test engineer embedded full-time, 1 software engineer for UI and integrations, 1 MLOps/platform engineer, and part-time quality and IT security representatives.

14. Risks and Mitigations

Risk	Impact	Mitigation
Poor genealogy or missing ECID	Weak RCA and FT-to-sort correlation	Phase 0 data audit; probabilistic joins with flagged uncertainty; push for ECID on new designs
Alarm fatigue	Engineers ignore the system	FDR control, risk-adjusted ranking, daily digest, per-user tuning
Spurious correlation in RCA	Wrong tool blamed, wasted effort	Commonality plus model agreement, confounder adjustment, 'association' labeling, experiment suggestions
Test removal causes escapes	Customer quality incident	Rule-of-three bound, audit sampling, staged rollout, automatic revert on any downstream rise
Model drift after program change	Silent accuracy loss	Program revision as an explicit feature and drift trigger; shadow re-validation
LLM hallucination	Misleading explanation	Tool-sourced numbers only, citation required, refusal on low evidence, audit log
Change resistance	Low adoption	Embedded test engineer, transparent evidence, start with time-saving analytics before recommendations

15. Appendix

15.1 Glossary

Term	Meaning
ATE	Automatic test equipment
CP / FT / SLT	Chip probe (wafer sort) / final test / system-level test
DPAT	Dynamic part average testing: outlier limits recomputed per wafer or lot
DPPM	Defective parts per million shipped
ECID	Electronic chip ID fused into each die
GDBN	Good die in bad neighborhood
GDPW	Gross die per wafer
NNR	Nearest neighbor residual
Overkill / underkill	Good die rejected / bad die shipped
PAT	Part average testing (AEC-Q001)
SHAP	Shapley additive explanations for model attribution
STDF	Standard Test Data Format (V4) used by ATE
WAT / PCM	Wafer acceptance test / process control monitor e-test

15.2 Pareto view used in the cockpit

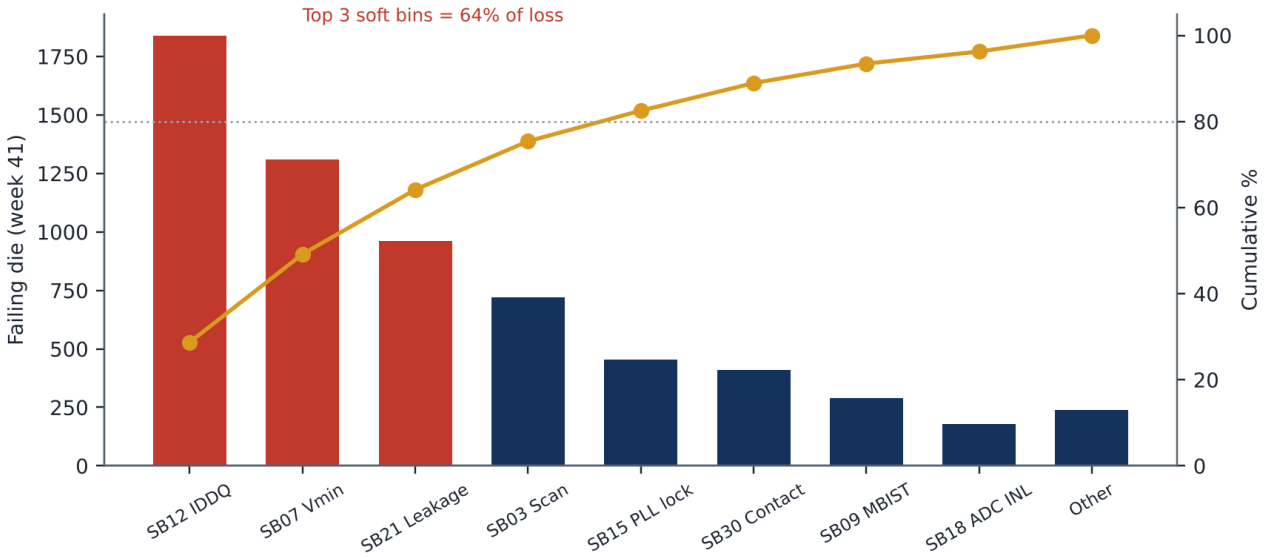


Figure 7. Soft-bin Pareto for one week. The Advisor links each top bin to its signature, root cause ranking, and open recommendations.

15.3 References

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